

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
CSE471	DEEP LEARNING FOR COMPUTER VISION	2	0	2	3	
Course Weightage	ATT: 5 CA: 45 ETT: 50	Exam Category: X1: Mid Term Exam: Not Applicable – End Term Exam: All MCQ				

Course Outcomes :Through this course students should be able to

CO1 :: remember the fundamental concepts of computer vision, image representation, preprocessing operations, and neural network basics used in deep learning.

CO2 :: understand the working principles of convolutional neural networks, feature extraction, pooling, regularization, and CNN training techniques.

CO3 :: apply transfer learning and fine-tuning methods using pre-trained models for domain-specific computer vision tasks.

CO4 :: apply object detection and localization techniques, including bounding box representation, IoU, NMS, and modern detection architectures.

CO5 :: apply image segmentation methods and transformer-based vision architectures for extracting meaningful visual information from images.

CO6 :: understand the principles of generative vision models, vision-language models, and interpretability techniques used in advanced computer vision applications.

	TextBooks (T)		
Sr No	Title	Author	Publisher Name
T-1	FUNDAMENTALS OF DEEP LEARNING & COMPUTER VISION	NIKHIL SINGH, PARAS AHUJA	BPB PUBLICATIONS

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	DEEP LEARNING	AMIT KUMAR DAS, SAPTRISHI GOSWAMI, PABITRA MITTRA, AMLAN CHAKRABARTI	Pearson Education India
R-2	ELEMENTS OF DEEP LEARNING FOR COMPUTER VISION	BHARAT SIKKA	BPB PUBLICATIONS
R-3	DEEP LEARNING FOR COMPUTER VISION: EXPERT TECHNIQUES TO TRAIN ADVANCED NEURAL NETWORKS USING TENSORFLOW AND KERAS	RAJALINGAPPAA SHANMUGAMANI	PACKT PUBLISHING

Other Reading (OR)	
Sr No	Journals articles as Compulsary reading (specific articles, complete reference)
OR-1	https://link.springer.com/article/10.1007/s00521-025-11827-w ,

Relevant Websites (RW)		
Sr No	(Web address) (only if relevant to the course)	Salient Features
RW-1	https://www.geeksforgeeks.org/computer-vision/difference-between-traditional-computer-vision-techniques-and-deep-learning-based-approaches/	Difference between Traditional Computer Vision Techniques and Deep Learning-based Approaches
RW-2	https://www.geeksforgeeks.org/deep-learning/convolutional-neural-network-cnn-in-machine-learning/	Convolutional Neural Network (CNN) in Deep Learning
RW-3	https://www.mygreatlearning.com/blog/what-is-transfer-learning/	What is Transfer Learning? Types and Applications
RW-4	https://labeleyourdata.com/articles/llm-fine-tuning/transfer-learning-vs-fine-tuning	Transfer Learning vs Fine Tuning: Key Differences for ML Engineers
RW-5	https://www.geeksforgeeks.org/computer-vision/what-is-object-detection-in-computer-vision/	What is Object Detection in Computer Vision?
RW-6	https://medium.com/@amitkharche/object-detection-models-explained-r-cnn-yolo-ssd-49b607c9ef6d	Object Detection Models Explained: R-CNN, YOLO, SSD
RW-7	https://huggingface.co/learn/computer-vision-course/en/unit3/vision-transformers/vision-transformers-for-image-segmentation	Transformer-based image segmentation
RW-8	https://link.springer.com/article/10.1007/s10462-024-10900-5	Transformers-based architectures for stroke segmentation: a review
RW-9	https://www.geeksforgeeks.org/deep-learning/generative-models-in-ai-a-comprehensive-comparison-of-gans-and-vaes/	Generative Models in AI: A Comprehensive Comparison of GANs and VAEs

Audio Visual Aids (AV)		
Sr No	(AV aids) (only if relevant to the course)	Salient Features
AV-1	https://www.youtube.com/watch?v=lt21C9bLRsg	Role of Deep Learning in Computer Vision Applications
AV-2	https://www.youtube.com/watch?v=Y3nkuU_kSNI	Types of Generative Adversarial Networks GAN Basics of Deep Learning

LTP week distribution: (LTP Weeks)	
Weeks before MTE	7
Weeks After MTE	7
Spill Over (Lecture)	4

Detailed Plan For Lectures

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Week Number	Lecture Number	Broad Topic(Sub Topic)	Chapters/Sections of Text/reference books	Other Readings, Relevant Websites, Audio Visual Aids, software and Virtual Labs	Lecture Description	Learning Outcomes	Pedagogical Tool Demonstration/ Case Study / Images / animation / ppt etc. Planned	Live Examples
Week 1	Lecture 1	Foundations of Computer Vision and Deep Learning (Role of computer vision in modern AI applications)	T-1	AV-1	Lecture-0 (Introduction to the course) and Role of computer vision in modern AI applications	Students will understand the importance of deep learning for computer vision in modern AI systems	PPT	Object identification in traffic scenario, agriculture, health diagnosis applications etc.
	Lecture 2	Foundations of Computer Vision and Deep Learning (Traditional versus deep-learning-based vision approaches)	T-1	AV-1	Comparison between deep learning and traditional method, fundamentals of images	Student will understand the logic behind the adaptation of deep learning models	PPT and Audio/Video means	Automatic text generation, language translation, game playing etc.
		Foundations of Computer Vision and Deep Learning (image fundamentals)	T-1	AV-1	Comparison between deep learning and traditional method, fundamentals of images	Student will understand the logic behind the adaptation of deep learning models	PPT and Audio/Video means	Automatic text generation, language translation, game playing etc.
		Foundations of Computer Vision and Deep Learning (pixels, channels, color spaces)	T-1	AV-1	Comparison between deep learning and traditional method, fundamentals of images	Student will understand the logic behind the adaptation of deep learning models	PPT and Audio/Video means	Automatic text generation, language translation, game playing etc.
		Foundations of Computer Vision and Deep Learning (resizing, cropping and normalization)	T-1	AV-1	Comparison between deep learning and traditional method, fundamentals of images	Student will understand the logic behind the adaptation of deep learning models	PPT and Audio/Video means	Automatic text generation, language translation, game playing etc.
Week 2	Lecture 3	Foundations of Computer Vision and Deep Learning (neural network basics)	T-1 R-2	AV-1	image fundamentals, pixels, channels, color spaces, resizing, cropping and normalization	Student will understand fundamental concepts related to images	PPT & Video demonstration	Image enhancement and segmentation examples
		Foundations of Computer Vision and Deep Learning (perceptron)	T-1 R-2	AV-1	image fundamentals, pixels, channels, color spaces, resizing, cropping and normalization	Student will understand fundamental concepts related to images	PPT & Video demonstration	Image enhancement and segmentation examples

Week 2	Lecture 3	Foundations of Computer Vision and Deep Learning (feedforward networks)	T-1 R-2	AV-1	image fundamentals, pixels, channels, color spaces, resizing, cropping and normalization	Student will understand fundamental concepts related to images	PPT & Video demonstration	Image enhancement and segmentation examples
	Lecture 4	Foundations of Computer Vision and Deep Learning (activation functions, loss functions)	T-1 R-1 R-2	AV-2	activation function, loss functions and optimization methods	The students will be familiar with the basic concepts of activation and loss functions.	PPT & numerical examples related to activation functions	Video demo of activation functions will be presented to students
		Foundations of Computer Vision and Deep Learning (optimization methods)	T-1 R-1 R-2	AV-2	activation function, loss functions and optimization methods	The students will be familiar with the basic concepts of activation and loss functions.	PPT & numerical examples related to activation functions	Video demo of activation functions will be presented to students
Week 3	Lecture 5	Convolutional Neural Networks and Training Techniques(CNN Basics: convolutional neural networks)	T-1 R-1 R-3	OR-1	Fundamentals about CNN, filters and feature maps	Students will be able to use the concept of CNN architecture and its basics	PPT	applications like NLP, Medical images and action recognition
		Convolutional Neural Networks and Training Techniques(convolution operation)	T-1 R-1 R-3	OR-1	Fundamentals about CNN, filters and feature maps	Students will be able to use the concept of CNN architecture and its basics	PPT	applications like NLP, Medical images and action recognition
		Convolutional Neural Networks and Training Techniques(filters, feature maps)	T-1 R-1 R-3	OR-1	Fundamentals about CNN, filters and feature maps	Students will be able to use the concept of CNN architecture and its basics	PPT	applications like NLP, Medical images and action recognition
	Lecture 6	Convolutional Neural Networks and Training Techniques(padding, stride)	T-1 R-1	OR-1	padding, stride, pooling operations of CNN along with Modern CNN architectures	Student will learn the Modern architectures of CNN along with its basic operations	PPT and Video presentation	LeNet, AlexNet, VGG, ResNet, and Inception networks
		Convolutional Neural Networks and Training Techniques(pooling, and receptive fields)	T-1 R-1	OR-1	padding, stride, pooling operations of CNN along with Modern CNN architectures	Student will learn the Modern architectures of CNN along with its basic operations	PPT and Video presentation	LeNet, AlexNet, VGG, ResNet, and Inception networks
		Convolutional Neural Networks and Training Techniques(Modern CNN architectures: LeNet, AlexNet, VGG, ResNet, and Inception networks)	T-1 R-1	OR-1	padding, stride, pooling operations of CNN along with Modern CNN architectures	Student will learn the Modern architectures of CNN along with its basic operations	PPT and Video presentation	LeNet, AlexNet, VGG, ResNet, and Inception networks

Week 4	Lecture 7	Convolutional Neural Networks and Training Techniques(regularization techniques)	T-1 R-2	OR-1	batch normalization and dropout, data augmentation methods	Students will be familiar with data augmentation methods and regularization techniques	PPT	examples related to data augmentation techniques
		Convolutional Neural Networks and Training Techniques(batch normalization and dropout)	T-1 R-2	OR-1	batch normalization and dropout, data augmentation methods	Students will be familiar with data augmentation methods and regularization techniques	PPT	examples related to data augmentation techniques
		Convolutional Neural Networks and Training Techniques(data augmentation methods)	T-1 R-2	OR-1	batch normalization and dropout, data augmentation methods	Students will be familiar with data augmentation methods and regularization techniques	PPT	examples related to data augmentation techniques
	Lecture 8	Convolutional Neural Networks and Training Techniques(learning rate scheduling)	T-1 R-1	RW-3	Learning rate scheduling and weight initialization	Students will be able to understand the concept of learning rate and early stopping criteria	PPT with numerical examples	learning rate changes and weights initialization examples
		Convolutional Neural Networks and Training Techniques(weight initialization and early stopping)	T-1 R-1	OR-1	Learning rate scheduling and weight initialization	Students will be able to understand the concept of learning rate and early stopping criteria	PPT with numerical examples	learning rate changes and weights initialization examples
Week 5	Lecture 9				Test 1			
	Lecture 10	Transfer Learning and Fine-Tuning for Vision Tasks (Pre-trained architecture: transfer learning)	T-1 R-1	RW-3	Transfer Learning and applications	Students will be able to understand the concept and its use of transfer learning technique	PPT	Examples related to feature extraction and fine tuning
		Transfer Learning and Fine-Tuning for Vision Tasks (ResNet)	T-1 R-1	RW-3	Transfer Learning and applications	Students will be able to understand the concept and its use of transfer learning technique	PPT	Examples related to feature extraction and fine tuning
		Transfer Learning and Fine-Tuning for Vision Tasks (EfficientNet)	T-1 R-1	RW-3	Transfer Learning and applications	Students will be able to understand the concept and its use of transfer learning technique	PPT	Examples related to feature extraction and fine tuning

Week 6	Lecture 11	Transfer Learning and Fine-Tuning for Vision Tasks (MobileNet)	R-2 R-3	RW-3 RW-4	description about MobileNet and feature extraction techniques	Students will be able to understand the concept of different features and their extraction process	PPT and examples	feature extraction examples according to specific applications
		Transfer Learning and Fine-Tuning for Vision Tasks (feature extraction)	R-2 R-3	RW-3 RW-4	description about MobileNet and feature extraction techniques	Students will be able to understand the concept of different features and their extraction process	PPT and examples	feature extraction examples according to specific applications
	Lecture 12	Transfer Learning and Fine-Tuning for Vision Tasks (fine-tuning)	T-1 R-3	RW-4	fine tuning, freezing and unfreezing of layers	Students will be able to understand the concept of fine tuning and managing layers	PPT	Fine tuning examples
		Transfer Learning and Fine-Tuning for Vision Tasks (managing freezing and unfreezing of layers)	T-1 R-3	RW-4	fine tuning, freezing and unfreezing of layers	Students will be able to understand the concept of fine tuning and managing layers	PPT	Fine tuning examples
Week 7	Lecture 13	Transfer Learning and Fine-Tuning for Vision Tasks (applying layer-wise learning rates)	T-1 R-2	RW-4	Domain adaptation and dataset learning	Students will be able to apply the concept of dataset and domain adaptation	PPT	Domain adaptation concepts and examples
		Transfer Learning and Fine-Tuning for Vision Tasks (domain adaptation and dataset)	T-1 R-2	RW-4	Domain adaptation and dataset learning	Students will be able to apply the concept of dataset and domain adaptation	PPT	Domain adaptation concepts and examples
		SPILL OVER						
Week 7	Lecture 14				Spill Over			
		MID-TERM						
Week 8	Lecture 15	Object Detection and Localization(object detection: object detection principles ,algorithms)	R-2 R-3	RW-5	Object detection, bounding box representation	Student will be able to identify objects	PPT and real time examples	Object detection examples
		Object Detection and Localization(bounding box representation)	R-2 R-3	RW-5	Object detection, bounding box representation	Student will be able to identify objects	PPT and real time examples	object detection examples
		Object Detection and Localization(intersection over union)	R-2 R-3	RW-5	Object detection, bounding box representation	Student will be able to identify objects	PPT and real time examples	Object detection and bounding box examples
		Object Detection and Localization(non-maximum suppression)	R-2 R-3	RW-5	Object detection, bounding box representation	Student will be able to identify objects	PPT and real time examples	Object detection and bounding box examples

Week 8	Lecture 16	Object Detection and Localization(classical detection pipelines)	R-1 R-2	RW-6	Classical detection pipeline, RCNN and single stage detector	student will be able to use classical detection pipeline and RCNN concept	PPT	idea about classification pipeline and its examples
		Object Detection and Localization(RCNN)	R-1 R-2	RW-6	Classical detection pipeline, RCNN and single stage detector	student will be able to use classical detection pipeline and RCNN concept	PPT	idea about classification pipeline and its examples
		Object Detection and Localization(single-stage detectors)	R-1 R-2	RW-6	Classical detection pipeline, RCNN and single stage detector	student will be able to use classical detection pipeline and RCNN concept	PPT	idea about classification pipeline and its examples
Week 9	Lecture 17	Object Detection and Localization(YOLO variants)	T-1 R-2	RW-6	YOLO, SD, RetinaNet with focal loss	Student will be able to understand YOLO variant	PTT, Video	YOLO and SD
		Object Detection and Localization(SSD, RetinaNet with focal loss)	T-1 R-2	RW-6	YOLO, SD, RetinaNet with focal loss	Student will be able to understand YOLO variant	PTT, Video	YOLO and SD
		Object Detection and Localization(Evaluation metrics: mean average precision, IoU thresholds)	T-1 R-2	RW-6	YOLO, SD, RetinaNet with focal loss	Student will be able to understand YOLO variant	PTT, Video	YOLO and SD
	Lecture 18	Image Segmentation and Advanced Vision Architectures(Image Segmentation: semantic segmentation)	T-1 R-1 R-2	RW-7	image segmentation and types, UNet	Students will be able to understand the task of segmentation	PPL, Video	image segmentation examples
		Image Segmentation and Advanced Vision Architectures(instance segmentation)	T-1 R-1 R-2	RW-7	image segmentation and types, UNet	Students will be able to understand the task of segmentation	PPL, Video	Image segmentation basid
		Image Segmentation and Advanced Vision Architectures(UNet)	T-1 R-1 R-2	RW-8	image segmentation and types, UNet	Students will be able to understand the task of segmentation	PPL, Video	Segmentation applications
Week 10	Lecture 19	Image Segmentation and Advanced Vision Architectures(DeepLab architectures)	T-1 R-2	RW-7	Segmentation using R-CNN and vision transformers	Students will be able to segment image data as per requirement	PTT, Video demonstration	Segmentation examples
		Image Segmentation and Advanced Vision Architectures(instance segmentation using Mask R-CNN)	T-1 R-2	RW-7	Segmentation using R-CNN and vision transformers	Students will be able to segment image data as per requirement	PTT, Video demonstration	Segmentation examples

Week 10	Lecture 19	Image Segmentation and Advanced Vision Architectures(Vision transformers: self-attention)	T-1 R-2	RW-7	Segmentation using R-CNN and vision transformers	Students will be able to segment image data as per requirement	PTT, Video demonstration	Segmentation examples
	Lecture 20	Image Segmentation and Advanced Vision Architectures(multi-head attention)	R-1 R-2	RW-8	Transformer based models , Positional encoding	Student will be able to user transformer models	PPT	positional encoding numerical examples
		Image Segmentation and Advanced Vision Architectures(positional encoding)	R-1 R-2	RW-8	Transformer based models , Positional encoding	Student will be able to user transformer models	PPT	positional encoding numerical examples
		Image Segmentation and Advanced Vision Architectures(transformer-based models, ViT, Swin Transformer)	R-1 R-2	RW-8	Transformer based models , Positional encoding	Student will be able to user transformer models	PPT	positional encoding numerical examples
Week 11	Lecture 21				Project			
	Lecture 22	Generative Vision Models (Generative deep learning: variational autoencoders)	T-1 R-3	RW-9	Generative deep learning	Student will be able to use generative deep learning	PPT	current examples of deep learning
		Generative Vision Models (GAN architectures)	T-1 R-3	RW-9	Generative deep learning	Student will be able to use generative deep learning	PPT	current examples of deep learning
Week 12	Lecture 23				Test 2			
	Lecture 24	Generative Vision Models (DCGAN)	R-1 R-2	RW-9	DGAN	Student will be able to understand the concept of DGAN and its use	PPT	examples related to DGAN
		Generative Vision Models (CycleGAN, StyleGAN)	R-1 R-2	RW-9	DGAN	Student will be able to understand the concept of DGAN and its use	PPT	examples related to DGAN
Week 13	Lecture 25	Generative Vision Models (image generation)	R-1	RW-1	Image generation	Students will be able to generate Images using these tools	PPT	examples of image generation basics
		Generative Vision Models (image-to-image translation, super-resolution)	R-1	RW-1	Image generation	Students will be able to generate Images using these tools	PPT	examples of image generation basics

Week 13	Lecture 26	Generative Vision Models (vision-language models: CLIP for image-text alignment)	T-1 R-2	RW-1 RW-2	vision-language models: CLIP for image-text alignment	Students will be able to understand the concept of vision transformers	PPT	examples related to vision-language models: CLIP for image-text alignment
		Generative Vision Models (interpretability techniques)	T-1 R-2	RW-1 RW-2	vision-language models: CLIP for image-text alignment	Students will be able to understand the concept of vision transformers	PPT	examples related to vision-language models: CLIP for image-text alignment
Week 14	Lecture 27	Generative Vision Models (Grad-CAM)	T-1 R-2	RW-2	saliency maps	Students will be able to learn the concept of saliency maps	PPT	examples related to saliency maps
		Generative Vision Models (saliency maps)	T-1 R-2	RW-2	saliency maps	Students will be able to learn the concept of saliency maps	PPT	examples related to saliency maps
		SPILL OVER						
Week 14	Lecture 28				Spill Over			
Week 15	Lecture 29				Spill Over			
	Lecture 30				Spill Over			

Scheme for CA:

CA Category of this Course Code is:C010102 (Total 3 tasks, 1 compulsory and out of remaining 1 best out of 2 to be considered)

Component	Iscompulsory	Weightage (%)	Mapped CO(s)
Test 1	NO	33.34	CO1, CO2
Project	Yes	66.66	CO1, CO2, CO3, CO4, CO5, CO6
Test 2	NO	33.34	CO1, CO2, CO3, CO4

Details of Academic Task(s)

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Project	The objective of this project is to apply complete deep learning for computer vision to solve real world problems by designing, implementing and evaluating the techniques.	Students are required to carry out a comprehensive computer vision project by covering the entire course syllabus, involving problem identification, dataset selection, text preprocessing, model design and implementation using deep learning and transformer-based approaches, performance evaluation using suitable metrics, and analysis of results, followed by proper documentation and presentation of the work.	Group	Online	30	3 / 11
Test 1	Student will be able to learn the basic concepts of computer vision and deep learning along with Convolution Neural Network architecture	Unit-I and Unit-II will be covered	Individual	Offline	30	4 / 5
Test 2	To test the subject knowledge of the students and provide opportunity to improve their CA performance in case of low marks or any missed CA	Subjective test covering syllabus from Unit-1 to Unit-IV	Individual	Offline	30	11 / 12

MOOCs/ Certification etc. mapped with the Academic Task(s)

Academic Task	Name Of Certification/Online Course/Test/Competition mapped	Type	Offered By Organisation
Test 1	DEEP LEARNING FOR COMPUTER VISION, IIT HYDERABAD BY PROF. VINEETH N BALASUBRAMANIAN	MOOCs	NPTEL

Where MOOCs/ Certification etc. are mapped with Academic Tasks:

- Students have choice to appear for Academic Task or MOOCs etc.
- The student may appear for both, In this case best obtained marks will be considered.

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
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Practical 1	List of Practical's	1. Load an image using OpenCV, perform resizing, grayscale conversion, normalization, and visualize the results using Matplotlib in multiple subplots.	AV-1	Students will be able to apply image operations like resizing, grayscale conversion, plotting graphs etc.
Practical 2	List of Practical's	2. Implement a 2D convolution operation using NumPy, apply it to a grayscale image patch with different filters, and visualize the resulting feature maps.	RW-2 AV-1	Students will learn the use of CNN model
Practical 3	List of Practical's	3. Implement a 2D convolution operation using NumPy, apply it to a grayscale image patch with different filters, and visualize the resulting feature maps.	RW-5	Students will be able to visualize feature maps using NumPy
Practical 4	List of Practical's	4. Train LeNet and ResNet-18 on MNIST and compare training time, accuracy, and model size to understand how architectural depth and skip connections influence performance.	RW-6	Student will be able to use LeNet and ResNet
Practical 5	List of Practical's	5. Use a pretrained to extract image features and train a classifier on a new dataset such as CIFAR-10 or a small custom dataset (e.g., flower images). Evaluate the classifier's accuracy.	RW-5	Student will be able to train a new classifier with desired features
Practical 6	List of Practical's	6. Fine-tune a pretrained MobileNet model on a small dataset of Cats vs Dogs. Experiment with freezing/unfreezing layers and observe the effect on training time and accuracy.	RW-5 RW-6	Student will be able to finetune the existing models.
Practical 7	List of Practical's	7. Use a pretrained YOLO model to detect objects in images or webcam/video streams, draw bounding boxes and class labels, and analyze detection confidence and performance	RW-6	Students will be able to use YOLO model effectively
Practical 8	List of Practical's	8. Fine-tune a pretrained Faster R-CNN model on a small custom dataset . Train for a few epochs and evaluate performance.	RW-1 RW-6	Students will be able to use faster RCNN model to evaluate the performance
Practical 9	List of Practical's	9. Use a pretrained UNet model to generate pixel-wise segmentation masks for sample images, overlay the predicted mask on the original image, and analyze the model's ability to segment objects	RW-2	Students will be able to use Pre-trained UNet model

Practical 10	List of Practical's	10. Use a pretrained CNN model to classify images, then apply Grad-CAM to visualize which parts of the image contributed most to the prediction. Compare Grad-CAM outputs for correctly classified and misclassified samples.		Students will be able to use Grad-CAM to visualize the parts of the image which contributed most
	SPILL OVER			
Practical 11	Spill Over			